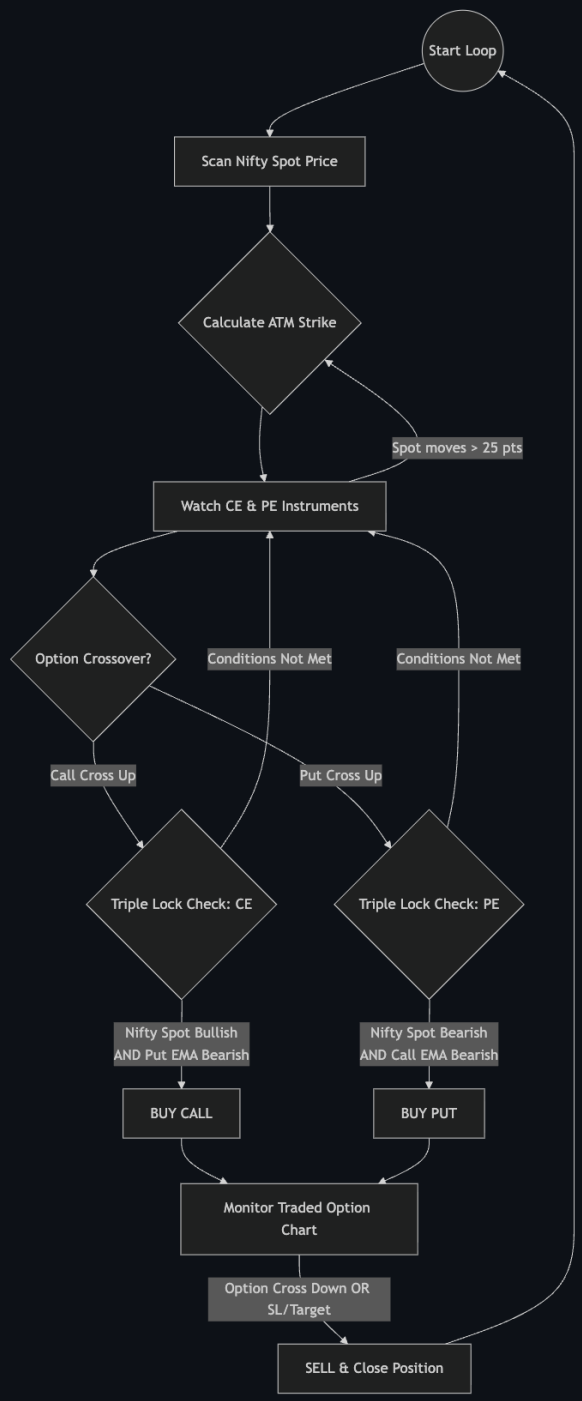
1. The Triple-Lock State Machine Diagram

Copy the code block below. If your AI tool supports Mermaid, it will visualize this perfectly.



2. Comprehensive Strategy Specification

Module: Triple-Lock TradeFlow Logic

A. Objective

Synchronize three independent data streams to filter out noise and capture high-velocity momentum. Entry is based on the Option chart; Confirmation is based on the Spot and Inverse Option.

B. Data Requirements

Instrument	Role	Required Indicators
Nifty Spot	Trend Anchor	EMA (Fast), EMA (Slow)
ATM Call (CE)	Execution/Trigger	EMA (Fast), EMA (Slow)
ATM Put (PE)	Inverse Filter	EMA (Fast), EMA (Slow)

ATM Put (PE)	Inverse Filter	EMA (Fast), EMA (Slow)

C. The State Machine Logic

Step 1: The Selection State

- Triggered at every **Candle Close**.
- Calculate `ATM_Strike = round(Nifty_Spot / 50) * 50`.
- Maintain a `Selection_Price`. If `abs(Nifty_Spot - Selection_Price) > 25`, re-run selection.

Step 2: The Signal State

- **CALL Buy Signal:** If `CE_EMA_Fast` crosses above `CE_EMA_Slow`.
- **PUT Buy Signal:** If `PE_EMA_Fast` crosses above `PE_EMA_Slow`.

Step 3: The Confirmation State (The Triple Lock)

If a **CALL Buy Signal** occurs, verify:

1. `Nifty_Spot_EMA_Fast > Nifty_Spot_EMA_Slow` (Market is Bullish).
2. `PE_EMA_Fast < PE_EMA_Slow` (Put is actually decaying/dropping). *All three must be true in the same/sequential candle window to trigger* `PositionManager.execute()`.

Step 4: The Exit State

- Once a trade is live, ignore Nifty Spot and the Inverse Option.
- Monitor **only** the active Option's chart.
- **Trigger Exit:** If `Active_Option_EMA_Fast` crosses below `Active_Option_EMA_Slow`.